

Meeting Notes

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May 11, 2026

The question is why the evolving NTK helps at smaller widths, but not at larger widths.

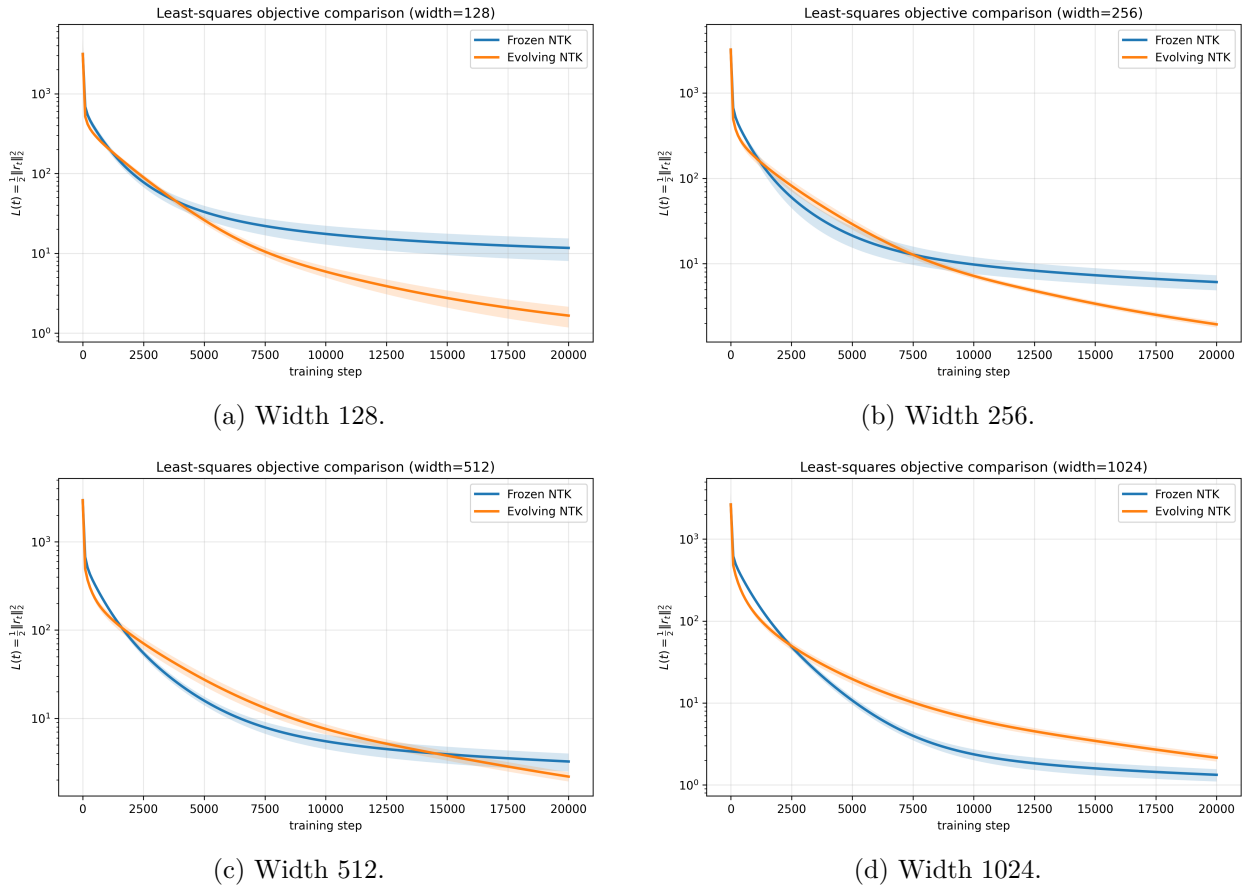


Figure 1: Least-squares objective comparison across widths. The evolving NTK gives a clear improvement at widths 128 and 256. The improvement is much smaller at width 512, and by width 1024 the frozen NTK gives the lower final objective.

The basic pattern is a width-dependent crossover. At small width, allowing the kernel to evolve improves optimization. At large width, the frozen kernel is already strong enough, and evolving the kernel no longer helps. To understand this, it is natural to first ask whether the evolving dynamics changes the Fourier subspaces themselves.

We therefore start with the projection error

$$\|\hat{P}_k(t) - P_k\|_F,$$

which measures how far the learned projector for frequency k moves away from the reference Fourier projector.

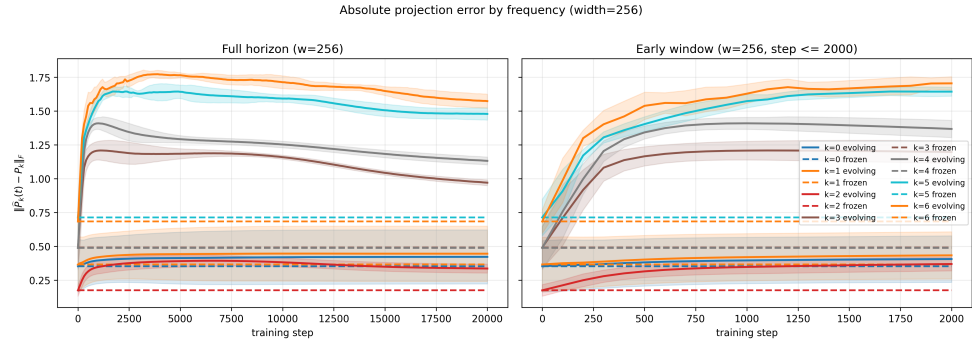
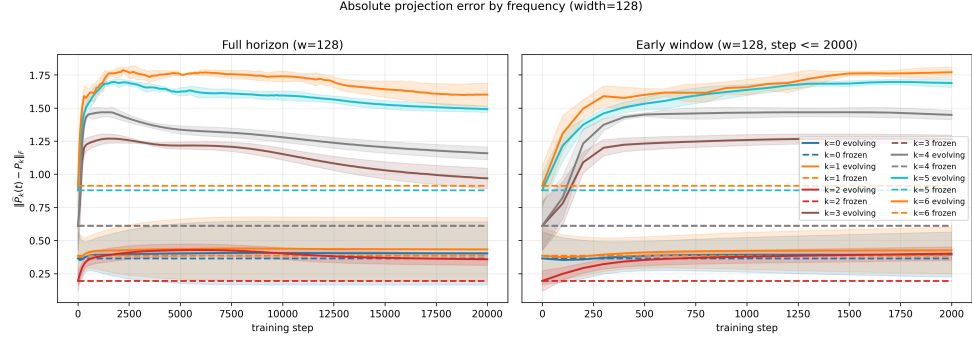
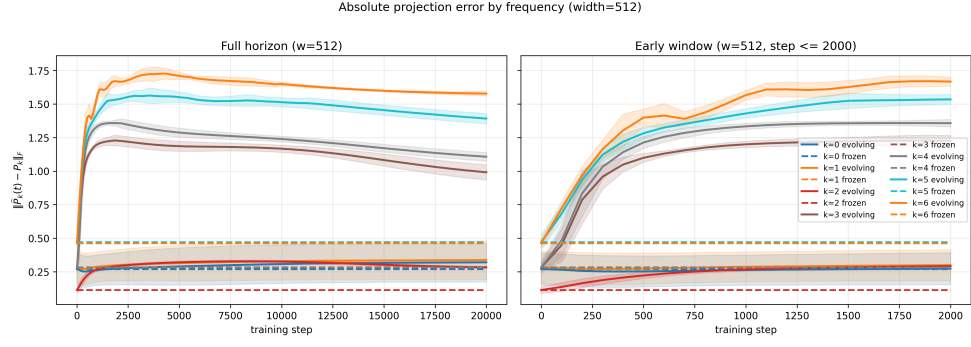
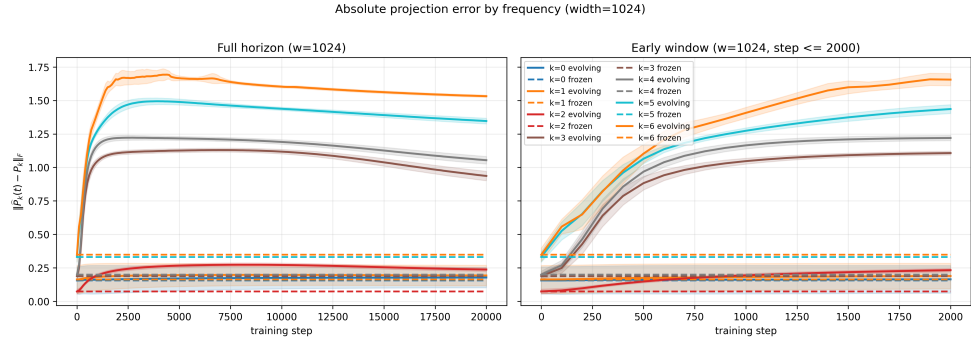


Figure 2: Absolute projection error by Fourier subspace for widths 128 and 256.

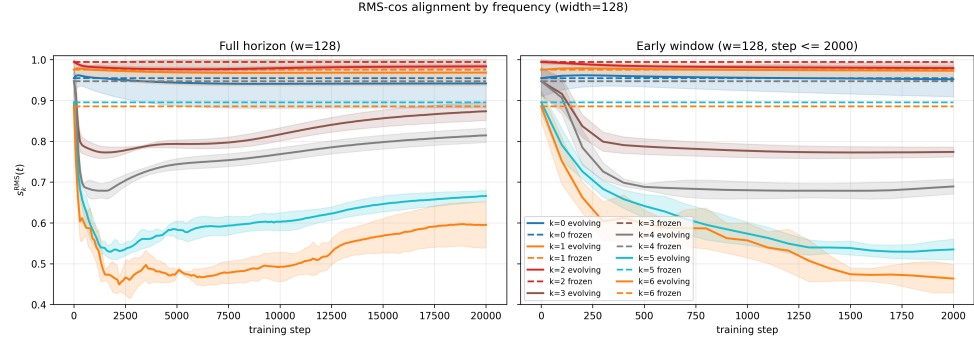


(a) Width 512.

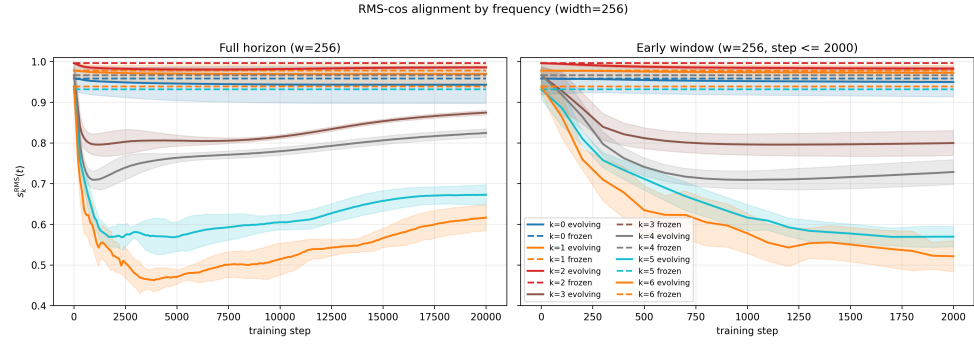


(b) Width 1024.

Figure 3: Absolute projection error by Fourier subspace for widths 512 and 1024.



(a) Width 128.



(b) Width 256.

Figure 4: RMS-cos alignment by Fourier subspace for widths 128 and 256.

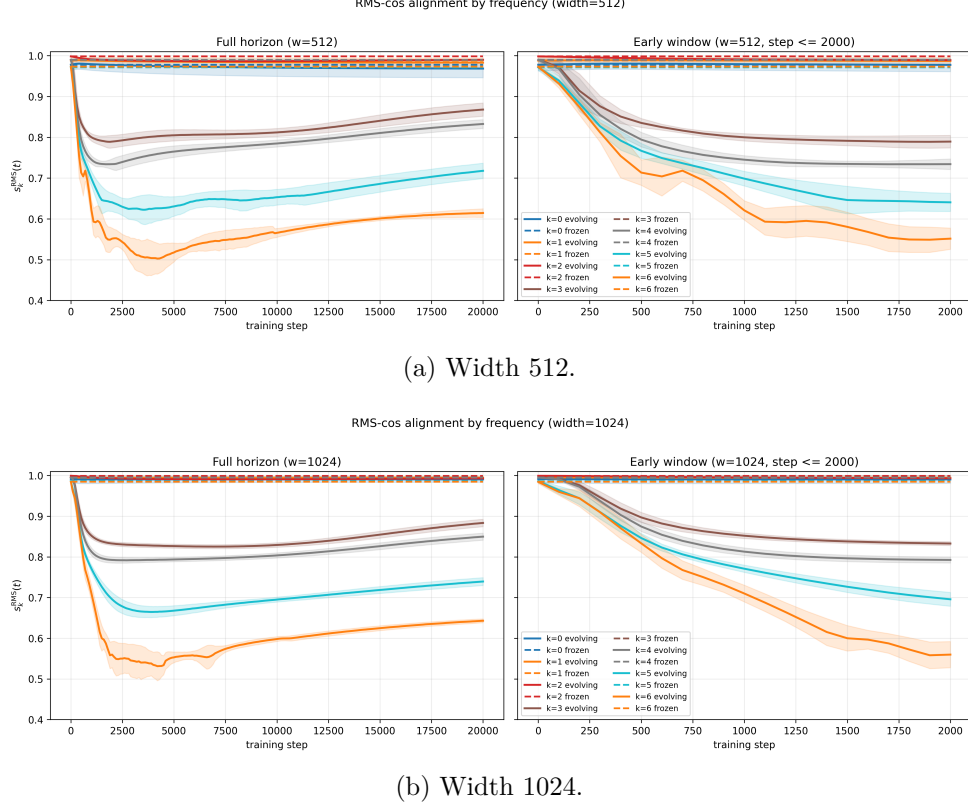


Figure 5: RMS-cos alignment by Fourier subspace for widths 512 and 1024.

The projection-error and cosine alignment plots show a consistent geometric trend. Under the evolving dynamics, several higher-frequency subspaces move away from their Fourier reference subspaces. The projection-error plots show this as an increase in $\|\hat{P}_k(t) - P_k\|_F$, while the cos alignment plots show it as a drop in subspace alignment.

This rules out a simple explanation based on improved Fourier geometry. The evolving NTK does not help because it becomes more Fourier-aligned. If anything, it becomes less Fourier-aligned, especially for higher frequencies. The main difference across widths is the starting point. At low width, the frozen NTK already starts with worse Fourier geometry; at high width, it starts much closer to the Fourier reference. Thus, at high width, feature learning moves the kernel away from a better initial geometry.

Therefore, projection error and cosine alignment should be read mainly as geometry diagnostics. They show that feature learning changes the Fourier subspaces, but they do not by themselves explain why the evolving model reaches lower loss at small width. To understand the loss, we also need to know how strongly the kernel acts in the directions being fitted. This is the distinction between geometry and strength.

A small toy example makes this point clearer. Let

$$\mathcal{F}_1 = \text{span}\{e_1, e_2\}$$

be a Fourier subspace, and let e_3 represent another frequency. Rotating the basis inside $\mathcal{F}_1$ does not change the subspace, and therefore does not change the projector P_1 . Projection error appears only when the subspace itself tilts into another frequency direction. For example, suppose the learned subspace is

$$\hat{\mathcal{F}}_1 = \text{span}\{\cos \theta e_1 + \sin \theta e_3, e_2\}.$$

Then $\hat{\mathcal{F}}_1 \neq \mathcal{F}_1$, and

$$\|\hat{P}_1 - P_1\|_F = \sqrt{2} \sin \theta.$$

So this is genuinely worse Fourier alignment: the learned frequency block is no longer purely the original Fourier block.

However, worse alignment does not automatically mean slower loss decay. To see this, consider the two-dimensional slice $\text{span}\{e_1, e_3\}$, where e_1 is a Fourier direction and e_3 represents a direction outside the original Fourier subspace. Suppose the residual is mainly in the Fourier direction

$$r = e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

A frozen, Fourier-aligned kernel may act on this slice as

$$K_{\text{frozen}} = \begin{pmatrix} 1 & 0 \\ 0 & 0.1 \end{pmatrix}.$$

Then

$$\langle e_1, K_{\text{frozen}} e_1 \rangle = 1.$$

Now suppose the evolving kernel tilts the leading eigendirection away from e_1 and toward e_3 . Let

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

The first tilted eigenvector is

$$u_1 = \cos \theta e_1 + \sin \theta e_3.$$

This is not a rotation inside the same Fourier subspace. It is a tilt into another frequency direction, so it would increase projection error.

Assume that the evolving kernel is diagonal in this tilted eigenbasis, with a larger leading eigenvalue:

$$K_{\text{evolving}} = R_\theta \begin{pmatrix} 3 & 0 \\ 0 & 0.1 \end{pmatrix} R_\theta^\top.$$

In the original (e_1, e_3) basis, this is

$$K_{\text{evolving}} = \begin{pmatrix} 3 \cos^2 \theta + 0.1 \sin^2 \theta & (3 - 0.1) \sin \theta \cos \theta \\ (3 - 0.1) \sin \theta \cos \theta & 3 \sin^2 \theta + 0.1 \cos^2 \theta \end{pmatrix}.$$

So for the same residual $r = e_1$,

$$\langle e_1, K_{\text{evolving}} e_1 \rangle = 3 \cos^2 \theta + 0.1 \sin^2 \theta.$$

For $\theta = 30^\circ$, this gives

$$3 \cdot 0.75 + 0.1 \cdot 0.25 = 2.275,$$

which is larger than the frozen value 1. Thus the evolving kernel can be less Fourier-aligned and still remove this residual direction faster, because the larger eigenvalue more than compensates for the tilt.

The example is not meant to prove that this is exactly what happens in the experiment. It only explains why the geometry plots are incomplete. Projection error and RMS-cos alignment measure whether the Fourier subspaces move. They do not measure whether the kernel becomes stronger

along the directions that still matter for the residual. This motivates the next diagnostic: spectral strength.

For each Fourier block $\mathcal{F}_k$, we summarize the average kernel strength by

$$\mu_{\mathcal{F}_k}(t) = \frac{1}{d_k} \text{tr}(P_k K_t P_k),$$

where P_k is the projector onto the k -th Fourier subspace and $d_k = \dim(\mathcal{F}_k)$. This is not an individual eigenvalue; it is the average spectral strength inside the subspace.

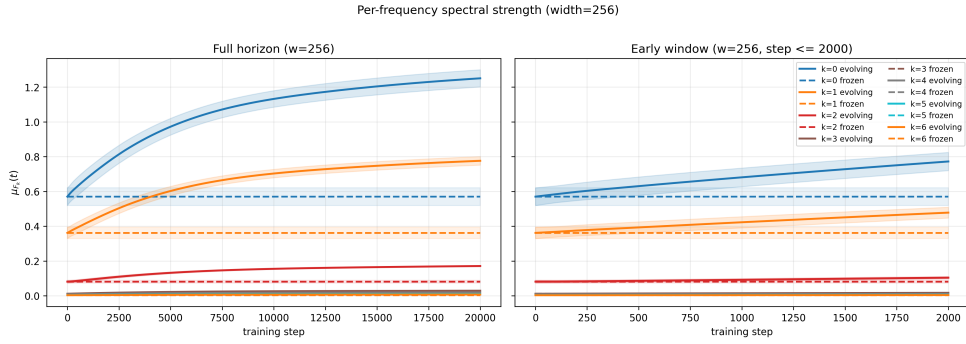
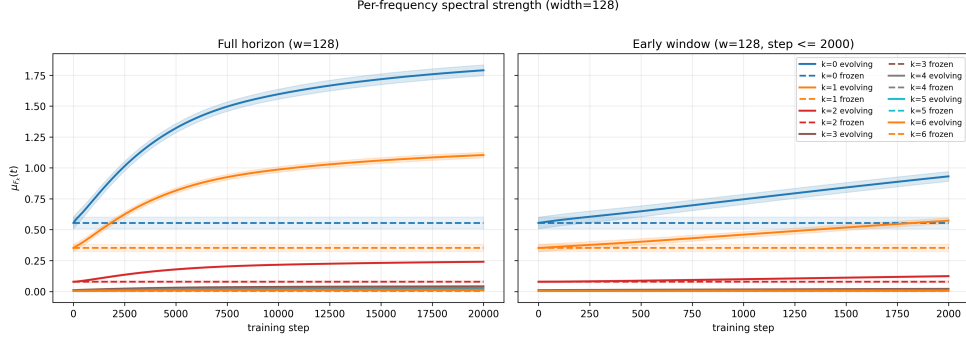
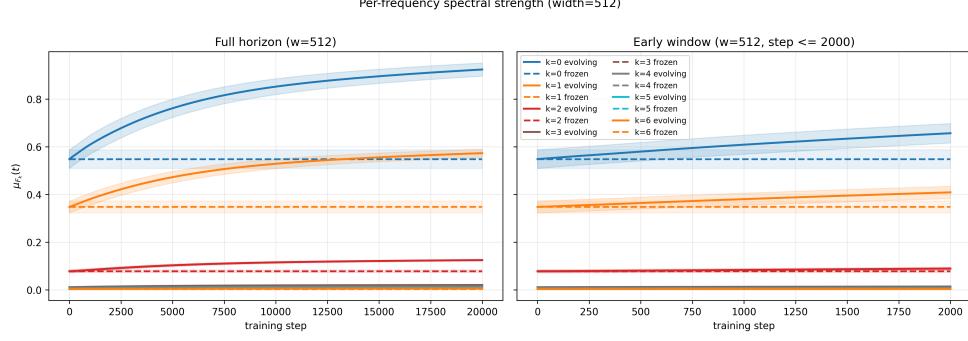
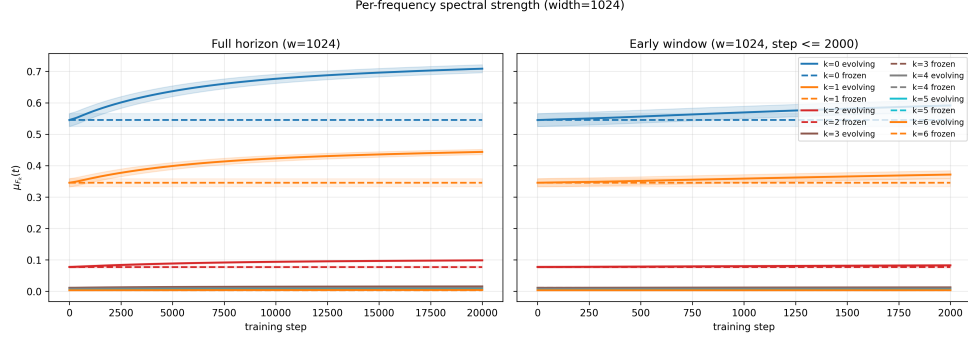


Figure 6: Per-frequency spectral strength for widths 128 and 256.



(a) Width 512.



(b) Width 1024.

Figure 7: Per-frequency spectral strength for widths 512 and 1024.

The spectral-strength plots give the missing part of the story. Across the different target-amplitude settings, the evolving NTK consistently increases the average kernel strength in the low-frequency blocks, especially $k = 0$ and $k = 1$, with a smaller increase for $k = 2$. The higher-frequency blocks remain much weaker. This is important because the same qualitative pattern appears even when the target amplitudes are changed: the low-frequency boost is not simply an artifact of assigning the largest amplitude to the constant or lowest-frequency mode.

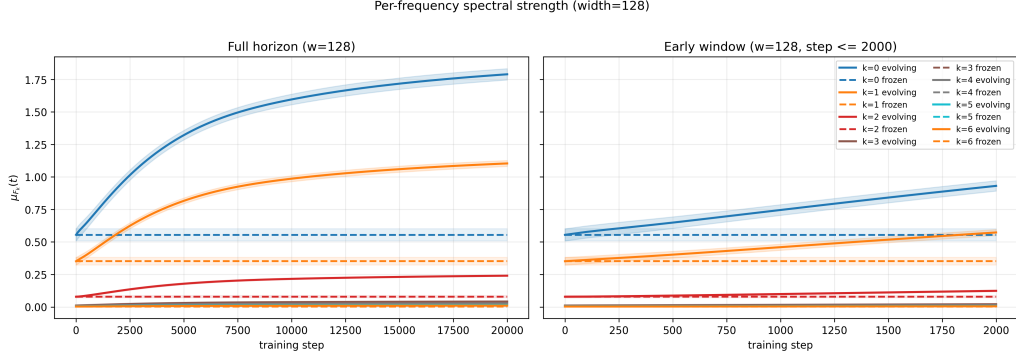
This suggests that the evolving kernel has a robust tendency to reweight spectral strength toward low frequencies. The geometry plots showed the cost: feature learning moves the Fourier subspaces and worsens alignment, especially for higher frequencies. The spectral-strength plots show the possible benefit: the same evolution also increases kernel strength in the low-frequency blocks. Thus the evolving NTK can improve the loss at small width despite worse Fourier geometry.

The width dependence is then naturally interpreted as a tradeoff. At low width, the frozen NTK is a rougher finite-width approximation, so the extra low-frequency strength gained through feature learning can outweigh the geometric drift. At high width, the frozen NTK starts closer to the Fourier reference and is already a strong baseline. The additional spectral reweighting from feature learning is then less useful relative to the cost of moving away from the Fourier geometry. This gives a plausible explanation for the crossover: feature learning helps when the low-frequency spectral gain is large enough to compensate for subspace drift, but loses its advantage once the frozen kernel is already stable and sufficiently strong.

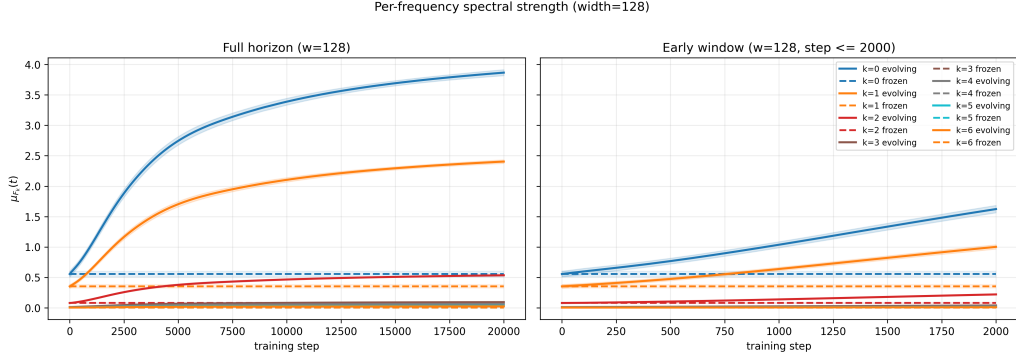
A concise takeaway is:

Feature learning does not help by preserving the Fourier geometry. The projection-error and cosine-alignment plots show the opposite: the evolving kernel moves the Fourier

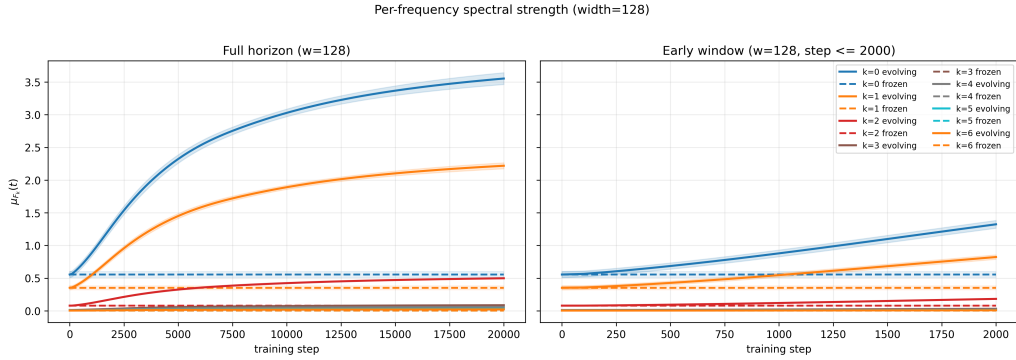
subspaces, especially at higher frequencies. The benefit appears to come from spectral reweighting instead. Across the amplitude variants, the evolving NTK increases low-frequency kernel strength, especially for $k = 0$ and $k = 1$. At small width, this gain can outweigh the geometric drift. At large width, the frozen NTK is already a good and stable approximation, so the extra gain from feature learning is no longer enough to compensate for the drift.



(a) Original amplitudes.

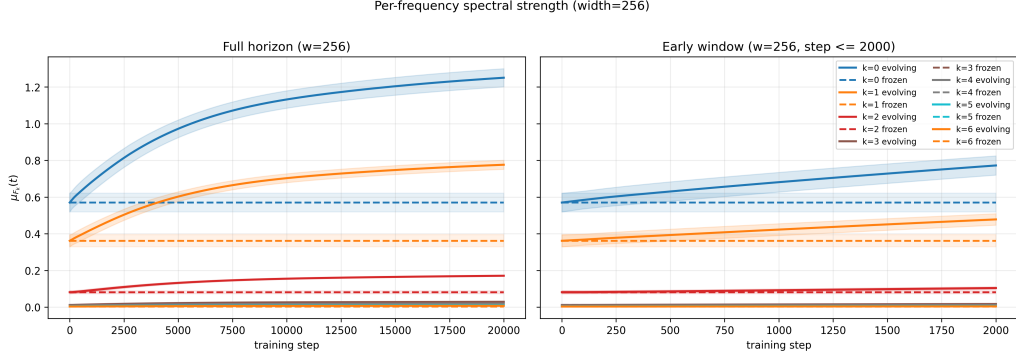


(b) Equal amplitudes.

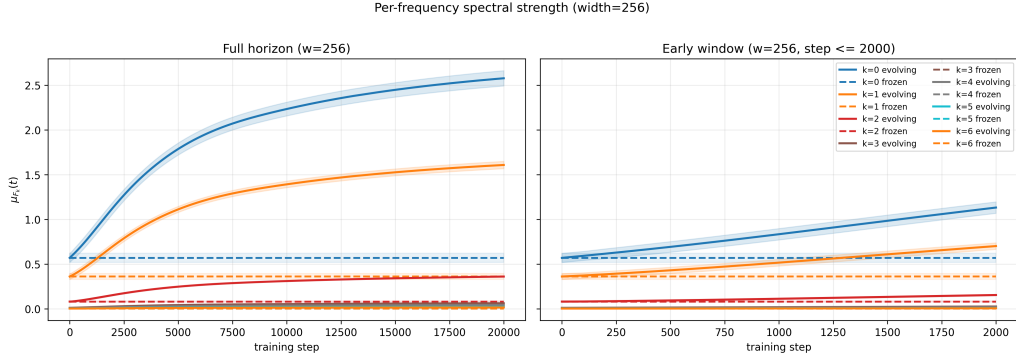


(c) Reversed amplitudes.

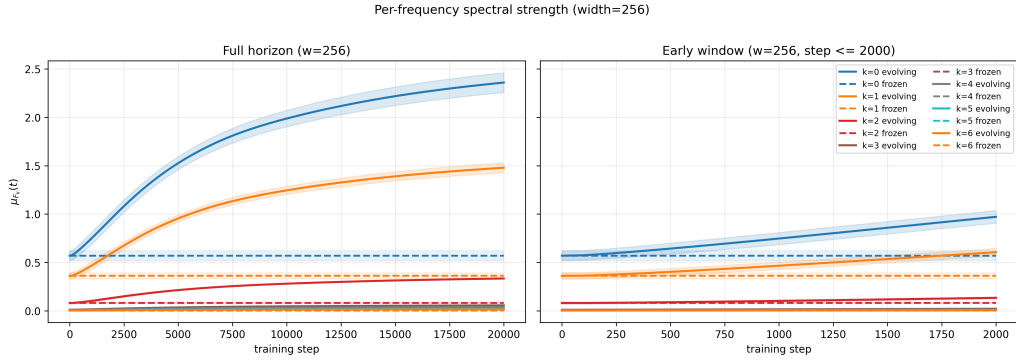
Figure 8: Per-frequency spectral strength at width 128 under the three target-amplitude settings.



(a) Original amplitudes.



(b) Equal amplitudes.



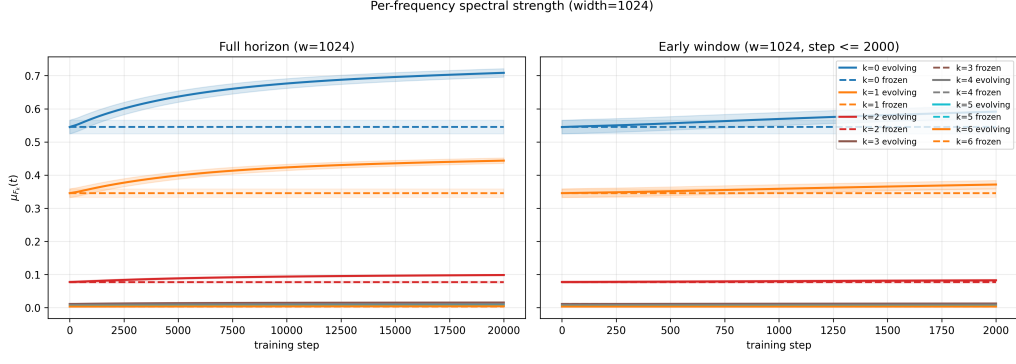
(c) Reversed amplitudes.

Figure 9: Per-frequency spectral strength at width 256 under the three target-amplitude settings.

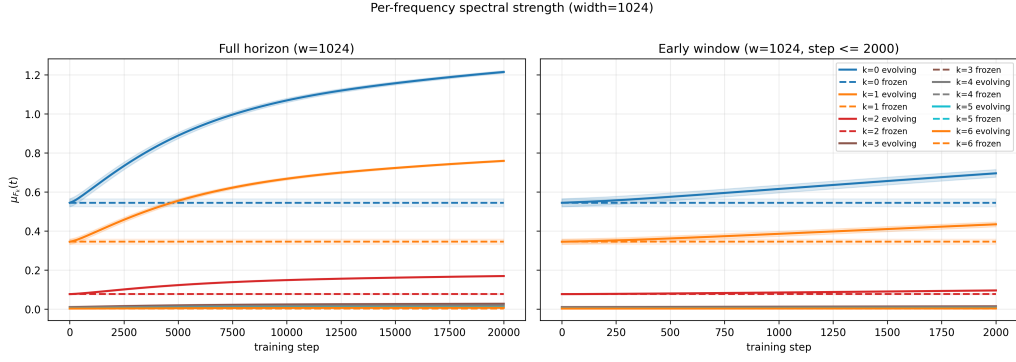
Figure 3 consists of two line plots side-by-side, showing the evolution of the mean $\mu_{k_i}(t)$ over training steps. The left plot is titled 'Full horizon (w=512)' and the right plot is titled 'Early window (w=512, step <= 2000)'. Both plots have 'training step' on the x-axis (0 to 20000 for the left, 0 to 2000 for the right) and ' $\mu_{k_i}(t)$ ' on the y-axis (0.0 to 0.8 for the left, 0.0 to 0.6 for the right). The legend indicates that solid lines represent 'evolving' models and dashed lines represent 'frozen' models for $k=0, 1, 2, 3, 4, 5, 6$. Shaded regions represent confidence intervals. In the full horizon plot, the $k=0$ evolving curve rises significantly above the frozen curves. In the early window plot, the curves are more stable and closer to their frozen counterparts.

Figure 10 consists of two line plots showing the per-frequency spectral strength $\mu_k(t)$ over training steps for different values of k (0 to 6). The left plot is for the 'Full horizon (w=512)' and the right plot is for the 'Early window (w=512, step ≤ 2000)'. In both plots, solid lines represent 'evolving' components and dashed lines represent 'frozen' components. Shaded regions indicate variance. In the full horizon plot, $k=0$ and $k=1$ components grow significantly, while in the early window plot, growth is much slower and limited to lower k values.

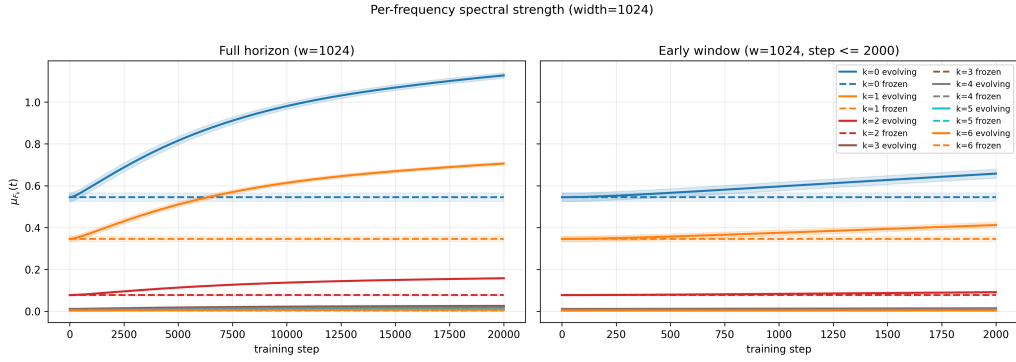
Figure 10: Per-frequency spectral strength at width 512 under the three target-amplitude settings.



(a) Original amplitudes.

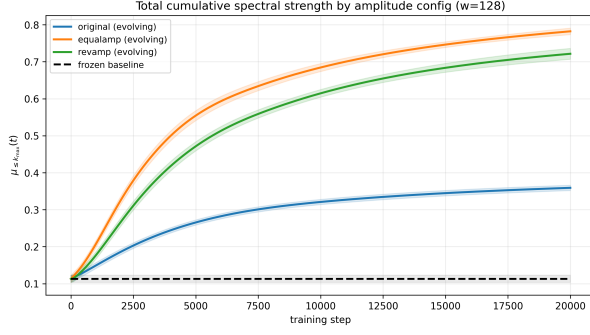


(b) Equal amplitudes.

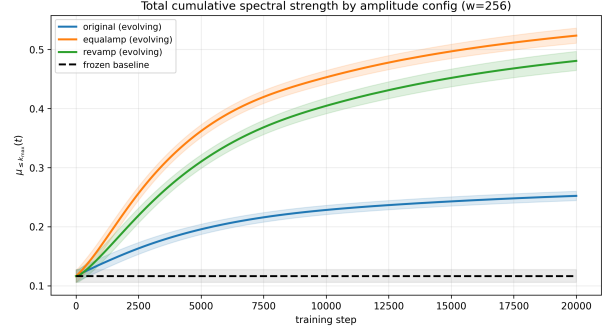


(c) Reversed amplitudes.

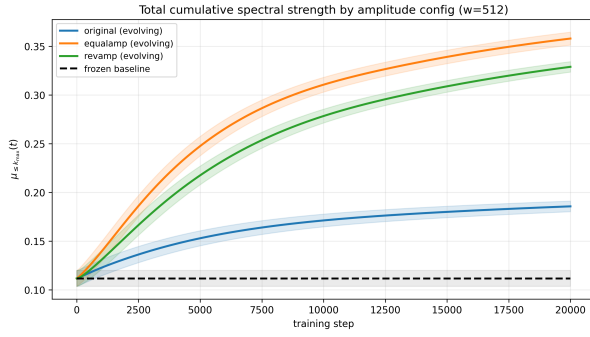
Figure 11: Per-frequency spectral strength at width 1024 under the three target-amplitude settings.



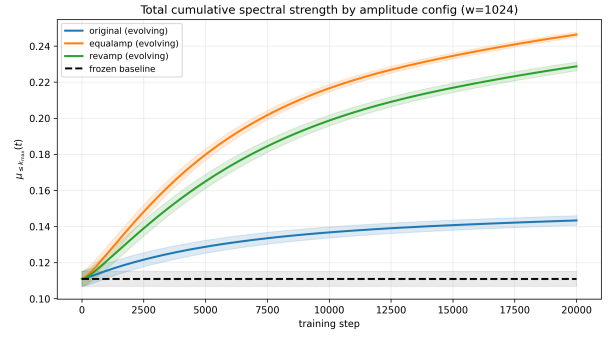
(a) Width 128.



(b) Width 256.



(c) Width 512.



(d) Width 1024.

Figure 12: Total cumulative spectral strength across amplitude configurations. The dashed line shows the frozen baseline, while the solid curves show the evolving NTK for the original, equal-amplitude, and reversed-amplitude targets.